

Bat Face Recognition: model $\times$ background sweep, ArcFace tuning, and interpretability

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1 Setup

Open-set bat face recognition on two species: **Mauritius** (16 identities) and **Rousettus** (12). Aligned 224px face crops (YOLO segmentation + pose $\rightarrow$ eye-levelled, mask-centred square crop) in three background variants: *original* (raw), *green* (chroma-key), and *random* (composited). Splits are identity-disjoint (Mauritius 12/2/2). Three model families: **Siamese** (4-conv, trained from scratch, 105px), **ArcFace** and **AdaFace** (ResNet50, ImageNet-pretrained, 112px). The dominant caveat throughout: the *test splits hold only 2–3 identities*, so single-run metrics are noise-heavy.

2 Settings tried

We ran the full $3 \times 3 \times 2$ matrix (model $\times$ background $\times$ species). Initial single-run results were unstable (the same ArcFace-green config gave ROC-AUC 0.910 then 0.623 on re-run), so we moved to **5-seed multi-seed** evaluation (deterministic, fixed data split; 90 runs) and report **mean $\pm$ std**. Everything is tracked in MLflow, tagged by **batch/species/seed**. Hyperparameter tuning used Optuna on the *validation* split only — the test split is never touched during selection.

3 ArcFace hyperparameters

The shared “tuned” recipe (Adam, lr=1e-3, margin $m = 0.2$, scale $s = 30$) worked for AdaFace but **ArcFace collapsed on Mauritius green/random** (single-run ROC-AUC ≈ 0.45 , below chance) while remaining fine on *original* (≈ 0.78). The stock ArcFace sweep searched $m \in [0.3, 0.6]$, $s \in [32, 96]$ — *above* the usable region: the paper-scale $m=0.5, s=64$ overpowers the angular space at ≤ 12 identities. With a corrected search space ($m \in [0.05, 0.4]$, $s \in [10, 48]$, lr $\in [2 \cdot 10^{-4}, 4 \cdot 10^{-3}]$) we re-tuned *per background*:

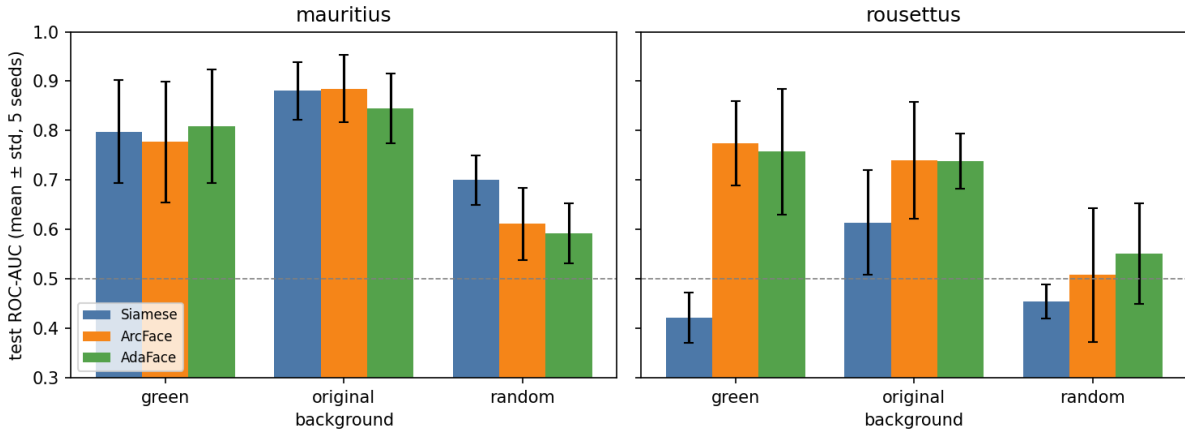
- **green**: $m=0.284$, $s=38.6$, lr= $3.2 \cdot 10^{-3}$;
- **random**: $m=0.186$, $s=37.6$, lr= $1.45 \cdot 10^{-3}$;
- **original**: kept the gentle shared recipe (the re-tune regressed it).

Higher scale (~ 38 vs 30) plus higher lr was the missing ingredient. *Caveat*: with only 2 validation identities, val-based selection is itself noisy (best-val trial 0.914 vs the re-run’s val 0.471); Optuna’s “best” is a starting point that must be confirmed by multi-seed test evaluation.

Model	Bg	Mauritius		Rousettus	
		ROC-AUC	top-1	ROC-AUC	top-1
Siamese	green	0.798 ± 0.105	—	0.421 ± 0.051	—
Siamese	original	0.880 ± 0.059	—	0.614 ± 0.106	—
Siamese	random	0.700 ± 0.050	—	0.454 ± 0.035	—
ArcFace	green	0.777 ± 0.122	0.741 ± 0.113	0.774 ± 0.085	0.700 ± 0.100
ArcFace	original	0.885 ± 0.068	0.900 ± 0.113	0.740 ± 0.119	0.796 ± 0.062
ArcFace	random	0.611 ± 0.073	0.582 ± 0.082	0.508 ± 0.135	0.485 ± 0.141
AdaFace	green	0.809 ± 0.114	0.794 ± 0.077	0.757 ± 0.127	0.667 ± 0.232
AdaFace	original	0.845 ± 0.071	0.841 ± 0.109	0.739 ± 0.056	0.759 ± 0.063
AdaFace	random	0.592 ± 0.061	0.662 ± 0.119	0.551 ± 0.102	0.542 ± 0.132

Table 1: Test metrics, 5-seed mean $\pm$ std (112px). Siamese reports no top-1 (verification-only pair model). ArcFace uses the per-background re-tuned recipe above.

4 Results



What the matrix says. (i) *Variance dominates*: per-cell std is ≈ 0.06 – 0.13 , so ROC-AUC gaps under ~ 0.14 are within seed noise — most within-cell model differences are statistical ties. (ii) Mauritius outperforms Rousettus overall; best cells are Mauritius ArcFace/Siamese *original* (≈ 0.88) and Rousettus ArcFace *green* (≈ 0.77). (iii) *Model ranking is species-dependent*: Siamese is strong on Mauritius (best on random, tied-best on original) yet collapses to $\sim$ chance on Rousettus, where the embedding heads (ArcFace/AdaFace) lead. (iv) The ArcFace re-tune made it *competitive* (no longer collapsing), not dominant — it ties AdaFace/Siamese within noise.

Input resolution. We retrained ArcFace/AdaFace at 224px (native crop size; ResNet50 is size-agnostic), 60 multi-seed runs. Higher resolution *did not help*: mean Δ ROC-AUC = -0.03 ; it improved only the hardest background (*random*, $+0.03$ avg) and degraded *green* (-0.06) and *original* (-0.07). (HPs were tuned at 112px, a partial confound.) Resolution is not the bottleneck — test-set size / seed variance is.

5 Conclusions

- On 2–3-identity test splits, seed variance (± 0.1 ROC-AUC) exceeds model, HP, and resolution differences — report mean $\pm$ std, never single runs.
- Best operating points: Mauritius ≈ 0.88 (original), Rousettus ≈ 0.77 (green/original, embedding heads). *Random* backgrounds are hardest everywhere.
- ArcFace needs background-specific margin/scale; the paper-scale defaults overpower small-identity sets. AdaFace is more robust to a shared recipe.
- More identities/data would move the needle more than higher resolution or head choice.

A Interpretability: where each model attends

Same bat per species (*original* background), maps at each model’s native resolution. **Siamese**: Integrated Gradients (20 steps) on the pair similarity. **ArcFace/AdaFace**: *SmoothGrad-IG* (denoised 20-step Integrated Gradients on the class logit, fine-grained) and *GradCAM* (last-conv class attention, smooth but coarse — ResNet50 at 112 px yields only a 4×4 grid). Each row shows the input, an overlay, and the attribution on black. The clean Siamese blobs (forehead/snout) reflect its shallow net; the ResNet50 pixel-gradients are inherently grainy, while GradCAM localises a smooth focus region on the face.

B Embedding structure: UMAP / t-SNE projections

ArcFace test-set embeddings projected to 2D (per-identity colour), *original* background. Tight, well-separated clusters indicate identity-discriminative embeddings; overlap flags the harder identities. Generated from the per-run projections via `gen_report_assets.py --mode projection`.

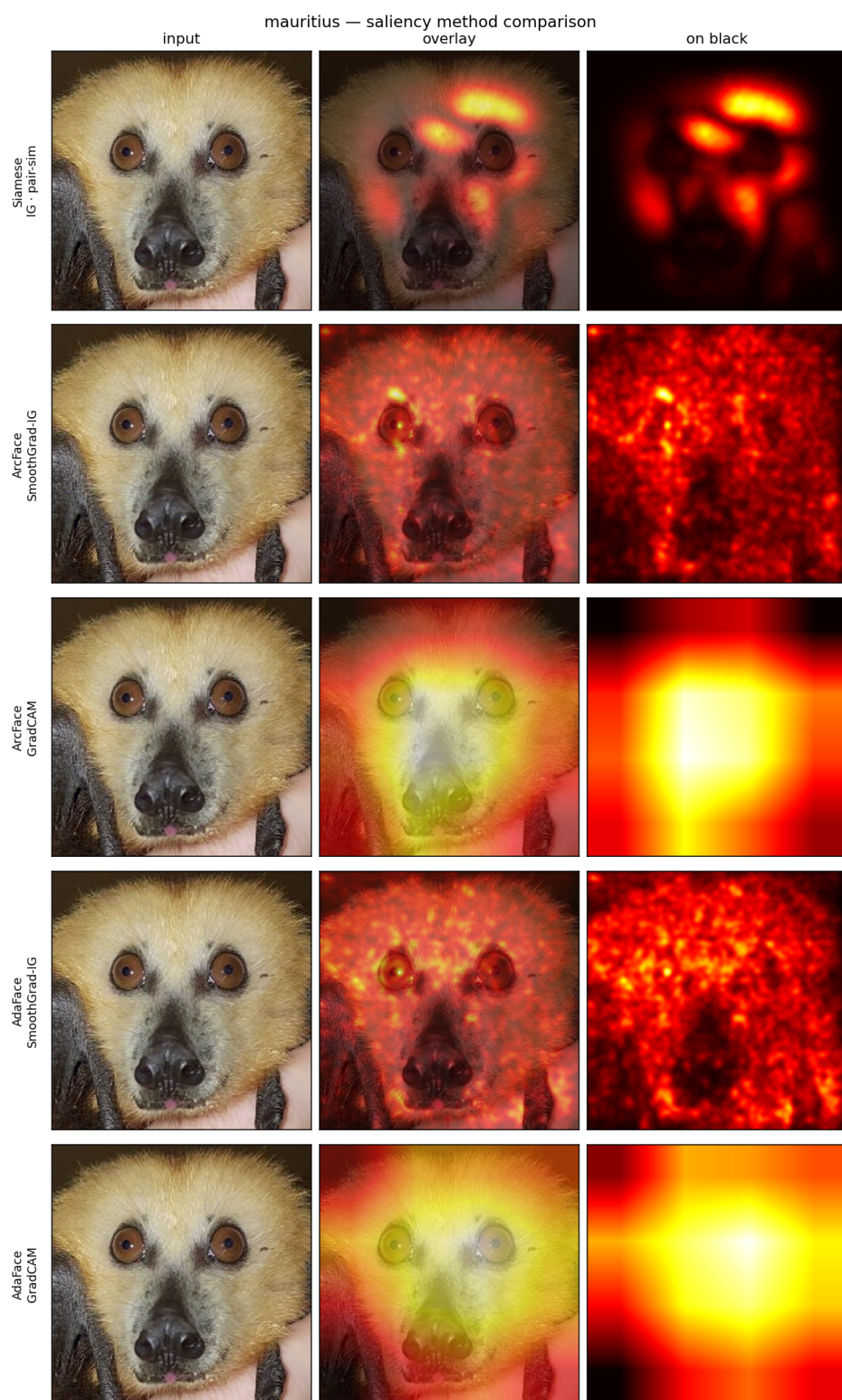


Figure 1: Mauritius — saliency across models/methods.

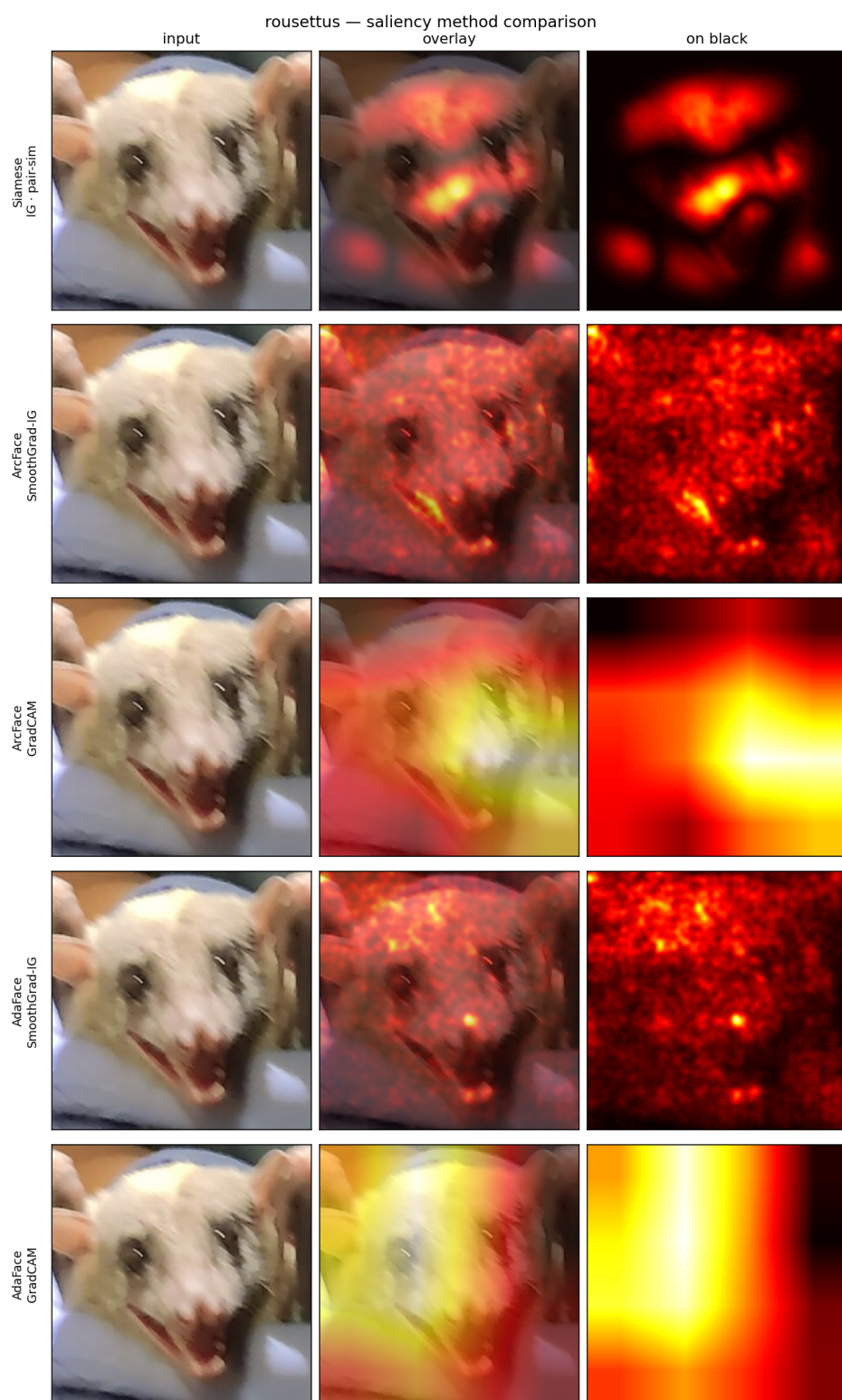


Figure 2: Rousettus — saliency across models/methods.

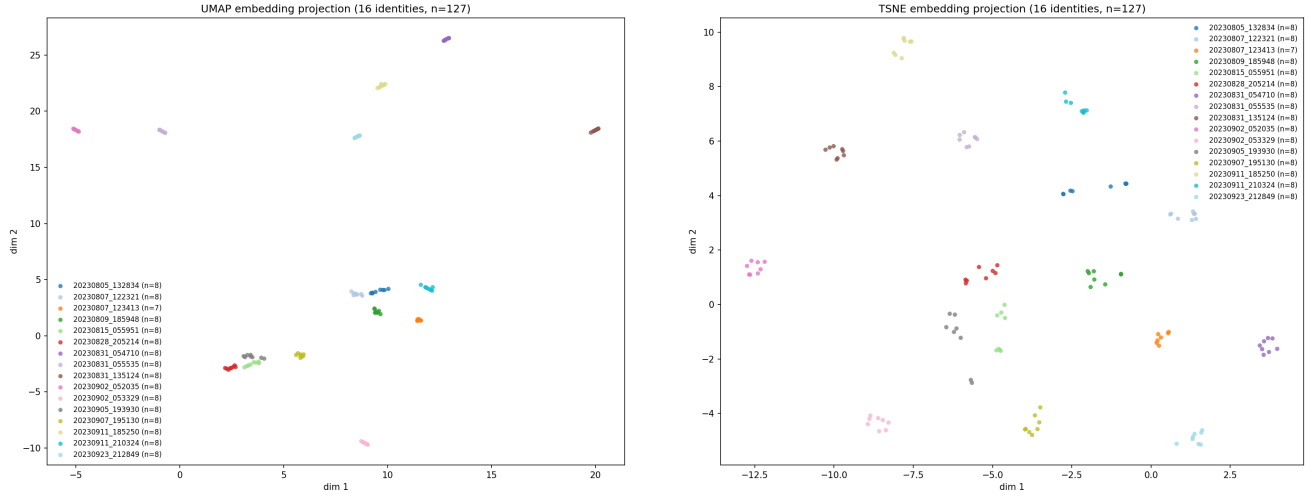


Figure 3: Mauritius — ArcFace embedding projections (UMAP left, t-SNE right).

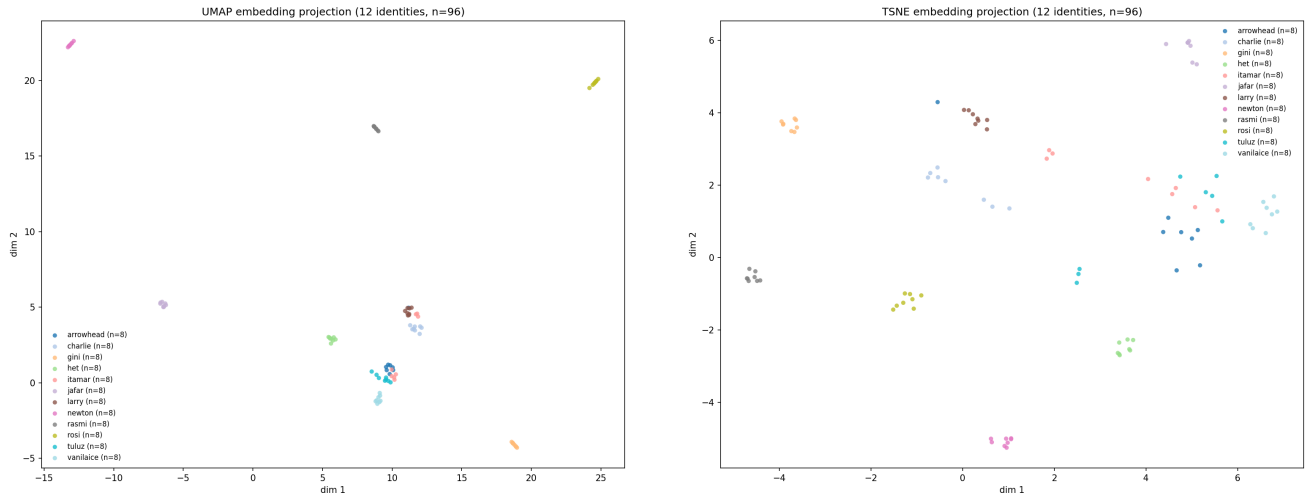


Figure 4: Rousettus — ArcFace embedding projections (UMAP left, t-SNE right).